

Gold-Free Faithfulness Metrics for NL-to-FOL Translation

Executive Summary

Internal Research Report

September 2026

1 Problem

Translating natural language into first-order logic (NL-FOL) is a prerequisite for neurosymbolic reasoning, yet evaluating whether a candidate formula faithfully captures a sentence’s meaning remains an open problem. Gold FOL annotations are expensive, non-unique, and unreliable: approximately 42% of entries in both FOLIO and MALLS contain incorrect formalisations. This investigation seeks gold-free metrics that predict faithfulness given only a sentence and a candidate formula, with no reference formula, no ontology, and no domain knowledge.

2 Approach

Over five iterations, we screened four metric families on a frozen evaluation set and a 700-sentence development set (E):

- **FOL-Triage:** a three-layer cascade (L1 formula lint, L2 bag-of-words text accounting, L3 role questionnaire with z3 formula reading).
- **Round-trip verification:** verbalise the formula back to natural language and compare via NLI.
- **Cross-system consensus:** collect independent translations from multiple LLM families, score by z3-equivalence agreement with the peer pool.
- **LLM judges:** direct scoring by cheap and frontier LLMs, with nonce disguise to control for memorisation.

The evaluation is a meta-evaluation: we build labelled (sentence, formula) pairs from audited public benchmarks and measure each metric’s AUROC for discriminating correct translations from errors. All confidence intervals are sentence-clustered percentile bootstrap ($B = 2000$, seed 0).

3 Key Findings

3.1 Cross-family consensus is the strongest signal

The cross-family consensus metric (c_score_align) achieves stratified AUROC 0.742 on development set E, beating the best cheap LLM judge (flash-lite disguised, 0.641) by +0.101 [+0.053, +0.145]. Three peer families suffice for 97% of full-pool performance.

On R_COMP SIG (controlled-vocabulary composed sentences, 221 sentences), consensus reaches within-template AUROC 0.954 with zero error endorsement, vs the judge’s 0.587 ($\Delta = +0.367$).

Adding `c_score_align` to a 28-feature baseline stack (`S4_full`) yields +0.039 [+0.021, +0.057] stratified AUROC. A 4-feature model (`c_score_align`, `L2-bow`, `L3`, `p_text`) matches the full stack without any judge features.

3.2 The rename tradeoff is unsolved

The consensus metric depends on shared predicate vocabulary between the candidate and its peers. Renaming predicates breaks the `z3` equivalence check:

Variant	RENAME_NONCE FA	SWAP recall
<code>c_align</code> (shipped V0)	0.765	0.758
<code>c_nf</code> (name-free)	0.132	0.512
<code>c_hyb</code> (hybrid)	0.216	0.832
GG (Gloss-Gated)	gate FAIL	—

No consensus variant achieves both low rename false-alarm rate and high structural-error recall. The GG checker (iteration 5) reduced `MEANING_RENAME` endorsement from 0.502 to 0.224 but failed the development gate (strat AUROC 0.696 vs V0 0.742).

3.3 Confirmation was blocked by budget

All three confirmation arms (E2-A: 110 sentences, E2-B: 400 long sentences, `R_COMP FREE`) were budget-stopped before reaching testability. The one testable cell—DT stratum on E2-A (179 ProverQA-derived rows with trusted gold)—yields V0 strat AUROC 0.916 [0.858, 0.970]. The L25 long-sentence stratum shows V0 near chance (0.443, 42 rows).

3.4 Dead ends

Nine approaches were tried and dropped:

Dead end	Iter	Key number
L1 lint on LLM outputs	1	AUROC 0.508
L2 role-aware	1	AUROC 0.557
Pilot structural metrics	1	AUROC 0.500–0.533
Round-trip reformalisation	1	AUROC 0.513
B2 decomposed <code>z3</code> reader	2	Gate BA 0.476
<code>R_ADJ</code> LLM adjudicator	2	Sonnet-5 BA 0.662
Name-free consensus (<code>c_nf</code>)	2–3	SWAP recall 0.512
CSC (Candidate-Signature)	4	Strat AUROC 0.614
GG (Gloss-Gated)	5	Strat AUROC 0.696

4 Limitations

1. **No confirmation.** Every confirmation arm was budget-stopped before reaching testability.
2. **Domain scope.** All evaluation is on FOLIO and MALLS-derived data; domain transfer is untested.
3. **Label noise.** Panel majority accuracy on expert-curated errors is 0.727; `R_COMP FREE` audit shows 13–28% label contamination.

4. **No complexity scaling.** The consensus advantage does not grow with sentence complexity (M3 INCONCLUSIVE).
5. **Rename tradeoff unsolved.** No variant achieves both low rename FA and high error recall.

5 Shipped Metric

V0 (`c_score.align`) stands as the shipped metric. No variant V1–V5 met the pre-registered selection rule (the best candidate, V4, gained only +0.0088 LONG-strat, below the +0.015 threshold).

Cost: \$0.000121 per item (consensus) vs \$0.0000488 per item (flash-lite judge). At $k = 3$ peer families: \$0.000946 per sentence.

Operating envelope: strongest on exception-clause sentences (AUROC 0.964 on “except” sentences) and on sentences with 12+ words where judges degrade. Weakest on short simple sentences (CTRL: consensus -0.069 vs judge, n.s.) and on vocabulary-mismatch errors (MEAN-ING_RENAME recall 0.07 at FA 0.10).

6 Total API Spend

Iteration	Spend
1 (screen)	\$13.66
2 (dev set E + PERTURB)	\$4.23
3 (API bar + R_COMP + PERTURB)	\$4.33
4 (CSC + E2 + FREE labels)	\$0.35
5 (E2-A/B + R_COMP FREE + freeze)	\$4.02
Total	\$26.59